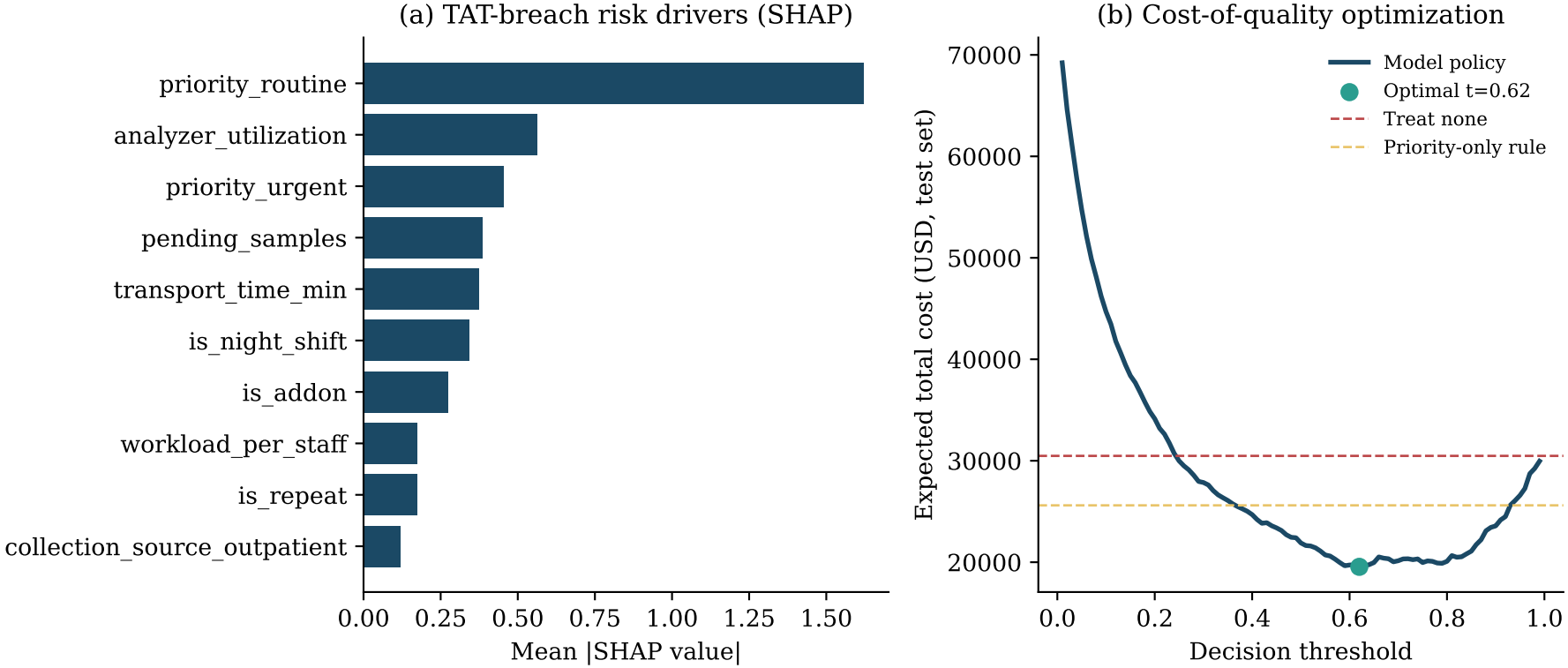
(a) TAT-breach risk drivers (SHAP)
priority_routine
analyzer_utilization
priority_urgent
pending_samples
transport_time_min
is_night_shift
is_addon
workload_per_staff
is_repeat
collection_source_outpatient
0.00
0.25
0.50
0.75
1.00
1.25
1.50
Mean |SHAP value|
(b) Cost-of-quality optimization
Model policy
Optimal t=0.62
Treat none
Priority-only rule
Expected total cost (USD, test set)
70000
60000
50000
40000
30000
20000
0.0
0.2
0.4
0.6
0.8
1.0
Decision threshold